

SMART SPREADSHEET - BRIEF APP REPORT

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1) What This App Is

Smart Spreadsheet is a multi-tenant SaaS analytics app for business teams that need quick insights from CSV exports without writing SQL.

It combines:

- CSV upload and dataset management
- Dashboard analytics (overview, trends, quality, KPI signals)
- AI-assisted Q&A and summaries
- Report generation and collaboration workflows
- India-focused insight mode for local business context

2) What The App Does Today

Core Data Workflow

1. User signs in (workspace + tenant context).
2. Uploads CSV data.
3. App computes metrics and analyst insights.
4. User asks business questions in natural language.
5. User generates reports and shares outputs.

Key Implemented Capabilities

- Auth and tenant-isolated workspaces ('/auth/*').
- Dataset ingest and retrieval ('/datasets/*').
- Overview metrics and analyst insights ('/overview/metrics').
- AI query/summarize flows ('/ai/query', '/ai/summarize').
- Data cleaning profile, preview, apply, rollback ('/cleaning/*').
- Report CRUD + sharing/comment/approval ('/reports/*').
- India insights and India trend reporting ('/india/*').
- Connector and workspace settings support.

3) Product Value

The app reduces time from "raw export" to "decision-ready summary" by combining deterministic analytics with assistant features in one workflow.

Primary value:

- Faster weekly business review cycles.
- Better visibility into margin/profit drivers.
- Clearer data quality issues before making decisions.
- Lower analytics complexity for non-technical teams.

4) Precision And Trust Status (Current)

The app uses deterministic backend calculations for core metrics and trends.

Recent precision upgrade includes:

- Improved parsing for messy numeric fields (currency symbols, commas, accounting negatives).
- Shared parsing logic used by overview analytics and cleaning engine.
- Precision audit metadata exposed in insights:
 - auto-coerced numeric columns
 - ignored numeric-like columns
 - coercion confidence score
- UI now shows "Precision Confidence" and precision notes.
- Automated test added for currency-formatted numeric parsing.

5) Tech Stack

- Frontend: Next.js + TypeScript
- Backend: FastAPI + SQLAlchemy + Pandas
- DB: SQLite (dev), PostgreSQL-ready production architecture
- Auth: JWT-based
- API proxying through Next.js backend route

6) Current Strengths

- Practical end-to-end analytics flow from upload to report.
- Multi-tenant structure ready for SaaS scaling.
- Deterministic insight layer reduces hallucination risk for core numbers.
- India-localized reporting makes the product region-friendly.
- Built-in cleaning and precision audit improve real-world usability.

7) Current Gaps

- Some advanced analytics remain heuristic for messy/unstructured schemas.
- Enterprise-grade controls can be expanded (deeper RBAC/audit depth).
- Production-scale rate limiting should move to distributed backing if needed.

8) Overall Assessment

Smart Spreadsheet is a solid, functional BI assistant product for SMB/mid-market operators. It is already useful for real reporting workflows and has clear differentiation through:

- combined cleaning + analytics + AI + report flow
- deterministic metric core
- India-focused insights mode

With continued UI polish and enterprise hardening, it is positioned to grow into a high-utility operational analytics platform.